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Fall 2024 MAT 2362
Instructor: Kirill Zaynullin

Midterm Test 1 (v1)

First name: _____ Last name: _____

Student number:

Please, read the following instructions carefully:

- This is a **closed book exam**. All electronic communication devices (e.g. cell phones, calculators, laptops, tablets) must be turned off and **put away in your bag**. Do not keep them in your possession, such as in your pockets. If caught with such a device (even if it is switched off), academic fraud allegations will be filed.
- Fill in your first name, the last name and the student number. **Do not detach** the pages of this examination.

This space is reserved for marker

Nr	*1*	*2*	*3*	*4*	*5*	*6*	Total
Mark							
Out of	4	3	2	3	4	4	20

1. Prove that for every integer $n > 6$ we have $n! > 2^n$. (4pt)

Proof:

2. Using truth tables, determine whether the following formula is a tautology, a contradiction, or none `(3pt)`

$$(X \Rightarrow Y) \vee (Y \Rightarrow X)$$

(correct answer without justifications is 0pts)

Proof:

3. For the following proposition write the **contrapositive** and the **converse**: (2pt)

If a series $\sum_{n=6}^{\infty} a_n$ converges, then the sequence $\{a_n\}_{n=6}^{\infty}$ converges and $\lim_{n\to\infty} a_n = 0$.

Answer:

4. Use the method of **proof by transposition** to prove that (3pt)

If ab is even then at least one of a and b is even.

Proof:

5. Prove that (4pt)

$$\forall_{x\in\mathbb{R}}\exists_{n\in\mathbb{N}}\ \frac{x^6+1}{n+1} < \frac{1}{19}.$$

Proof:

6. Define a relation $\sim$ on $\mathbb{R}^2$ by stating that $(a, b) \sim (c, d)$ if and only if $a^6 + b^6 \leq c^6 + d^6$.
Prove that $\sim$ is reflexive and transitive but not symmetric. (4pt)

Proof:

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